

Cache Translation Across Model Scale: An Audit with Identity Controls and Oracle Probes

immediate

Abstract

Cache translation maps the key-value (KV) cache that a large teacher forms over a context into the cache space of a smaller student, so that the student can answer without re-prefilling the context. Recent work reports quality-preserving translation across heterogeneous models; we audit that claim under strict zero re-prefill with three gated hypotheses and two instruments that prior evaluations omit. An *identity control* injects the student’s own cache through the full translation-and-injection pipeline and reproduces the student’s own prefill (per-sample logit cosine 0.99998; gold-probability change +0.0001, 95% CI $[-0.0006, +0.0009]$), separating pipeline mechanics from mapping; a cold-start run that loads only the persisted cache and mapper parameters reproduces the offline evaluation exactly. *Oracle probes* then mix teacher-cache content into the student’s own cache at controlled fractions and layer windows. Across seven mapper families, four scale pairs, and five alternative students, translation does not lift a strong student above its own prefill: point estimates lie between -0.392 and -0.138 , the calibration ladder on one fixed evaluation set stays within $[-0.257, -0.238]$ from 10 to 200 examples, and an Heo-inspired top- k /de-RoPE ablation under our audit regime—top- k cross-layer source selection with de-RoPE keys, in both its averaged and its concatenated FineWeb-Edu-calibrated form—is worse still (CHG -0.22 to -0.31 , every interval below zero). The only translated configurations whose intervals clear our reporting replacement margin ($\epsilon=0.02$) are two near-chance students (accuracy 0.30 and 0.27) whose gains are indistinguishable from zero. A 0.6B student is not established as non-inferior ($+0.010$ $[-0.081, +0.102]$ for 4B $\rightarrow$ 0.6B; -0.023 $[-0.104, +0.064]$ for 8B $\rightarrow$ 0.6B). No probe configuration yields a gain whose interval excludes zero, at any mixture fraction or layer window; teacher-native content degrades the student (-0.334 $[-0.508, -0.154]$); and a translator that restores near-native fluency (perplexity 23.7 versus 21.2) still leaves accuracy at 0.267 against the student’s 0.500. Held-out reconstruction R^2 is high for keys ($+0.81$) and partial for values ($+0.32$), and a larger position-wise joint mapper reconstructs both channels better (0.85 and 0.48) yet performs worse on the task—reconstruction fidelity does not imply capability. Under zero re-prefill, we find no evidence that the teacher’s answer-relevant advantage is recoverable from its cache by the tested translators. The numeric replacement margin $\epsilon=0.02$ was selected after the gate form was frozen and is therefore used as a practical reporting tolerance, not as a preregistered constant (Appendix A); the gain gate is ϵ -free and no configuration passes it. All 171 audit runs are recorded for reproducibility, and we release the protocol as a checklist for future claims.

1 Introduction

Multi-model serving systems want to reuse the teacher’s work. A large teacher prefills a long context once; a smaller student should inherit that state and answer at teacher quality without re-reading the context. Cache translation formalizes this idea: learn a map $\mathbf{KV}^T \mapsto \widehat{\mathbf{KV}}^S$ from the teacher’s cache space into the student’s. Recent systems report encouraging results, including Mixture-of-Translators (MoT), which preserves 51.0% closed-set QA accuracy at 7B $\rightarrow$ 0.5B scale [13], Cache-to-Cache semantic communication [5], latent-space communication [3], and selective KV sharing [17].

These reports conflate three questions that can fail independently. Does the injection pipeline consume a foreign cache correctly (*mechanics*)? Does the map replace the student’s own cache at replacement level (*mapping*)? Does the teacher’s cache carry student-readable content that lifts the student above its own prefill (*exploitability*)? Prior evaluations score the translated end-to-end pipeline against the student, so a positive delta could come from any of the three; a null result cannot be attributed. The line also lacks an

identity control, a probe of the exploitability ceiling, and any report of what happens to *accuracy* when perplexity is repaired.

We evaluate cache translation as an audit with three gated hypotheses. **H1 (injection mechanics)** asks whether the cache-injection pipeline preserves a valid student-space cache without loss; its instrument is an identity control that injects the student’s own cache through the full pipeline. **H2 (mapping)** asks whether any translator reaches replacement level; its instrument is a ladder of seven mapper families under a disjoint calibration ladder of 30 to 200 examples, including a residual-anchored translator (RAT) that we derive from the two models’ own projections and a shared vocabulary alignment. **H3 (exploitability)** asks whether the teacher’s cache contains any student-readable advantage at all; its instrument is an oracle probe that mixes teacher-cache content into the student’s own cache at controlled fractions and layer windows. Figure 1 shows the setting, the instruments, and the gates.

The audit returns three verdicts; Section 3.3 labels which of them were pre-specified and which are extensions made while revising. H1 passes: the identity control reproduces the student’s own prefill (per-sample logit cosine 0.99998; gold-probability change +0.0001, 95% CI $[-0.0006, +0.0009]$), so mechanics is not the failure mode. no tested strong-student mapper establishes replacement: gold-probability change over the student’s own prefill ranges from -0.138 (best configuration) to -0.392 (worst) across the mapper ladder, the best configuration reaches 0.365 against a 0.503 student baseline, and on one fixed evaluation set the calibration ladder stays within $[-0.257, -0.238]$. The weak 0.6B student is established as neither worse nor non-inferior (+0.010, CI $[-0.081, +0.102]$ at $n=63$); at the paired variance we observe, clearing the reporting margin would take roughly an order of magnitude more evaluation samples (Section 5). H3 returns a negative that we state narrowly: no configuration we test produces a gain whose interval excludes zero. The blended-probe point estimate falls monotonically with the teacher fraction ($0.503 \rightarrow 0.380$ at $\alpha=0.25$, 0.249 at $\alpha=0.75$) and separates from the student from $\alpha=0.5$ onward, while the top-layer windows are indistinguishable from the student, with point estimates up to +0.020 whose intervals span zero. A translator that restores near-native fluency (PPL 23.7 versus 21.2 for the identity control) still leaves accuracy at 0.267: a cache can read fluently and still miss the answer.

Three things fall out of the audit. The first is a framework: identity controls, disjoint calibration/evaluation splits, gold-probability metrics with bootstrap intervals and permutation tests, and oracle probes that test exploitability without training anything (Section 4). The second is a bounded design space: seven mapper families, a calibration ladder, and component ablations, all under one protocol (Section 4, Section 5); we additionally derive the most architecture-informed translator we could (RAT), and it fails too, which strengthens the negative. The third is a verdict with consequences: mechanics passes, no strong-student mapper establishes replacement and is inconclusive for the weak one, no probe configuration with teacher content produces an interval-excluded gain, and perplexity cannot stand in for accuracy as evidence (Section 5, Section 7). H2’s null result routes the audit to H3 because the probes measure what a frozen student can extract from teacher-cache content without training anything, so a null result cannot be blamed on the mapper.

2 A Survey of Cache-Level Communication Between Models

We organize prior work into four lines and close each with the assumption it leaves untested. Table 1 summarizes the comparison.

2.1 KV Reuse Within a Single Model

Prompt caching, cache-augmented generation, and streaming attention reuse a model’s own cache across requests. Streaming attention identifies attention sinks at early positions whose KV carries outsized attention mass [18]. CacheGen compresses and streams a KV cache across a network [14]; CacheBlend fuses several cached fragments with selective attention recomputation for RAG [20]; KVSharer shares caches across layers that behave dissimilarly [19]. These systems assume a single model on both sides of the cache. The reuse question is therefore about storage and attention cost, not about representation compatibility. Our audit inherits their serving motivation but moves the mismatch to the model axis, where the compatibility assumption breaks.

2.2 Cross-Model Cache Translation

The line closest to our audit translates caches between models. C2C proposes direct cache-to-cache semantic communication with a depth-ratio layer mapping [5]. LSC aligns caches through a shared latent space with a cross-attention translator [3]. MoT combines gated translator modules over depth-ratio channel windows with a Context Correction Loss that aligns a replayed target trajectory with the native one, and reports the strongest numbers to date: 51.0% average closed-set QA accuracy at Qwen2.5-7B→0.5B and 96.3% retention in long-context cache-augmented generation [13]. KVComm shares attention-ranked KV layers [17]; Interlat communicates last-hidden-state messages [4]; HCache restores caches from hidden-state translations [6]. Two properties of this line matter for our audit. First, its evaluations score the translated pipeline against the student baseline, which conflates mechanics, mapping, and exploitability. Second, its strongest variants do not operate under strict zero re-prefill: MoT reconstructs the target-side cache by replaying the context through the target model with source-guided sparse attention, so the reported quality includes a correction pass that a cold-start deployment would not get. A naive target-side replay pass does not by itself recover the strong student (Appendix C), so MoT’s gains cannot be attributed to replay alone. Our audit holds the zero-re-prefill constraint fixed and asks what a translator alone can deliver. Our numbers sit far below the 73–98% retention reported for closed-form cache transfer on matched-KV pairs [10]. The two quantities are not like for like: retention measures how closely a translated cache reproduces the receiver’s own prefill, not whether the receiver gains over it, and the calibration and evaluation splits differ. Other systems in this line additionally relax zero re-prefill: MoT reconstructs the target-side cache with source-guided sparse attention and a correction loss. We also implement the closest strict-zero-reprefill design: per-target-layer top- k source selection by held-out ridge reconstruction score, with de-RoPE keys, run at $c=200$ on the fixed evaluation set. It degrades the strong student further than our proportional-mapping ridge does (-0.27 to -0.31 ; every interval below zero, Table 2), so the deficit we measure is not explained by omitting Heo et al.’s layer-selection or key-alignment choices.

2.3 Cross-Model Representation Alignment

A separate line asks whether two trained networks’ representations can be aligned at all. Model stitching connects frozen networks with a learned linear layer and shows that stitching quality varies sharply across training runs and layers [1]. Relative representations replace absolute coordinates with anchor-based similarity spaces and enable zero-shot communication between encoders [15]. The Platonic Representation Hypothesis argues that representations converge across models up to linear transform [11], with refinements and proofs following [8, 21]. `vec2vec` translates embeddings between models without paired data by exploiting this shared geometry [12]. These results motivate the hypothesis that a linear map between cache spaces should exist. Our results qualify the conclusion for caches: the convergence evidence comes from pooled embedding spaces, while a cache is a per-layer, per-head, position-indexed state whose student-relevant content is not determined by its teacher counterpart (Section 6).

2.4 Frozen-Model Controllability

Work on steering shows that a frozen model’s behavior can be moved by injecting activations: task vectors compress in-context learning into a single mid-layer activation [9]; activation steering adds behavioral directions to the residual stream; gist tokens compress prompts into a few virtual tokens [16]. The tuned lens reads intermediate layers through per-layer affine probes [2], and the FFN-as-memory view assigns the factual capacity of a model to its feed-forward weights [7]. This line supports the intuition that a frozen student can absorb injected information, which makes the H3 null result informative rather than tautological: the failure is not that frozen students cannot be steered. It is that, in our tested teacher–student regime, raw KV translation does not deliver a usable steering signal to the frozen student. More transferable control signals may be easier to access in residual-stream directions or task-vector representations than in per-head KV entries.

Table 1: Cache-translation and adjacent work, organized by what is translated and what is held fixed. “Replay” marks methods whose reported quality includes a target-side correction pass over the context; our audit holds strict zero re-prefill. No cited evaluation isolates mechanics with an identity injection or reports accuracy at matched perplexity; MoT’s injection-window ablations are the closest antecedent (Section 2).

Work	What moves	Alignment	Replay	Unexamined assumption
C2C [5]	Full KV	depth-ratio lay- ers	no	a linear map per channel suffices
LSC [3]	KV via shared latent	learned latent space	no	one latent space fits all layers
MoT [13]	KV windows	gated translators	yes (sparse)	pipeline delta $\neq$ translation value
KVComm [17]	selected KV layers	attention ranking	no	ranked layers carry the signal
Interlat [4]	last hidden state	latent message	no	the last layer summarizes the cache
HCache [6]	hidden states	translator	no	states determine caches
Closed-form ridge [10]	full KV (ridge)	top- k layer selection, de-RoPE keys	no	retention $\neq$ gain; shared splits
This paper	full cache (zero re-prefill)	seven families incl. RAT	no	<i>identity-controlled evaluation</i> + diagnostic probes

2.5 What the Survey Does Not Establish

Across the four lines, no work (i) separates mechanics from mapping from exploitability, (ii) injects the student’s own cache as an identity control, (iii) reports accuracy at matched perplexity, or (iv) constrains exploitability with an oracle that requires no training. The claims of the translation line therefore rest on pipelines whose components have never been isolated. The audit below supplies the missing instruments.

3 The Three-Hypothesis Audit Framework

3.1 Setup

Let a teacher T prefill a context x and form a cache $C_T(x) = \{K_\ell^T, V_\ell^T\}_{\ell=1}^{L_T}$ with L_T layers, H KV heads, and head dimension d . A translator g_θ maps it to $\widehat{C}_S(x)$ in the space of a student S with L_S layers. The student then answers a query q with only the query tokens as input, $\hat{y} = S(q \mid \widehat{C}_S(x))$, under strict zero re-prefill: the context never enters the student’s forward pass. Three references anchor the audit. *Student self-prefill* $S(q \mid C_S(x))$ is the baseline the handoff must not degrade. *Identity injection* feeds $C_S(x)$ itself through the injection pipeline; it isolates mechanics from mapping. *Teacher full* $T(q \mid C_T(x))$ is the teacher full-prefill reference. We measure answer quality by the gold-letter probability p_{gold} —the softmax over the four answer-letter logits, $p_{\text{gold}} = e^{z_{c^*}} / \sum_{c \in \{A,B,C,D\}} e^{z_c}$, taken at the final prompt position with each model’s own tokenizer id for each letter (single-token letters, no leading space; only four-choice examples are retained), so teacher-side rows use the teacher’s letter tokens and student-side rows the student’s—and the change $\text{CHG} = p_{\text{gold}}(\text{handoff}) - p_{\text{gold}}(\text{student self})$, with 95% bootstrap intervals (Appendix A); permutation tests are also recorded but no reported decision uses a p -value; perplexity over the scoring suffix is recorded but never accepted as evidence of capability (Section 5.5).

3.2 Hypotheses, Instruments, Gates

Figure 1 states the three hypotheses with their instruments and gates. **H1**: the identity injection matches the student’s own prefill per sample (gate: mean logit cosine = 1 within float tolerance). **H2**: some g_θ

in the audited family reaches replacement level (gate: the CI lower bound of gold CHG is above a small tolerance $-\epsilon$; a stricter *gain* gate additionally requires the CI lower bound > 0). Replacement asks only that the student is not degraded; gain asks that the teacher’s cache provides a measurable lift. We report replacement results at $\epsilon = 0.02$ gold probability; the non-inferiority gate form was frozen before the audit runs, whereas this numeric tolerance was selected later as a practical reporting margin (Appendix A reports sensitivity to alternative margins), about 4% relative to the 0.503 student baseline—making the replacement gate a standard non-inferiority test of $H_0: \text{CHG} \leq -\epsilon$ against $H_1: \text{CHG} > -\epsilon$; unlike a naive significance test, it does not additionally require the point estimate to be non-negative. **H3**: the teacher’s cache contains student-readable advantage (gate: some oracle probe configuration with teacher content improves over the student’s own cache). H3’s probe does not depend on training a translator: a positive effect would demonstrate student-readable teacher-cache content directly, while a null result constrains the extraction paths we test without excluding every nonlinear one.

3.3 Evidence Tiers and Multiplicity

The audit grew during revision, and we do not let later additions inherit the authority of pre-specified tests. We therefore label every claim by how it entered the protocol. *Confirmatory* claims were fixed before the runs: the identity gate, the non-inferiority *form* of the replacement gate and the ϵ -free gain gate on the primary pair, and the fraction and window probes under the rule that exploitability is established only if some configuration’s interval excludes zero on the positive side. *Frozen-gate extensions* were added while revising but evaluated with the same gates and metric definitions: the residual-anchored and MLP translators, the 8B teacher, the calibration ladder, the cross-architecture and token-aligned students, the 1K-token retrieval task, the octant sweep, and the refit-variance study. We report those as robustness evidence rather than as independent confirmation. *Exploratory* analyses are the ones we derived from the recorded runs afterwards—the sample-size calculation, the reconstruction R^2 decomposition, and the mechanical reading of Section 6—and we word them as interpretations.

Breadth interacts with error rates in a direction that matters here. Our family of tests asks whether some configuration produces a *positive* effect; in the audit-protocol runs reported below, every point estimate is at or below $+0.010$ for a translation and at or below $+0.020$ for a diagnostic probe window, and no interval excludes zero on the positive side. A wider search increases the chance of finding a false positive if one exists, so searching more families, pairs, tasks, and budgets and still finding none reduces concern that the negative result is specific to one hand-picked configuration. We therefore treat the aggregate statement—no tested configuration produces a gain whose interval excludes zero—as the headline result, and report individual configurations with their own intervals rather than selecting among them. Earlier exploratory runs under a superseded protocol (confidence-based metric, in-sample calibration, and a cache-mutation bug) did produce small positive deltas; none survives the audit protocol, and we exclude them from every number in this paper.

For future *positive* uses of the same gate, the “some configuration is positive” rule must be evaluated with a family-wise correction or a max-statistic permutation test: per-configuration intervals at $\alpha=0.05$ would inflate the family-wise error rate across mixture fractions, layer windows, and content sources. In our data no positive interval occurs, so such a correction cannot create a positive finding here; it can only shrink one.

4 Audit Methodology

4.1 Identity Control

At evaluation time we capture the student’s own cache over the context (allowed in calibration and probe stages), feed it through the same mapper interface, cache construction, position handling, and scoring path as a translated cache, and score. Any deviation from the student’s own prefill is mechanics loss. This is the control that prior evaluations omit.

4.2 Disjoint Splits, Unified Prompts, and Honest Counters

Mapper calibration and evaluation draws are disjoint samples of the same distribution (HellaSwag and ARC-Challenge, shuffled then split). All methods share one token-level prompt format, so the only difference between

the handoff and the student baseline is where the context representation comes from. Zero-prefill counters read the actual cache length before and after the scoring forward pass rather than trusting bookkeeping, and the scoring pass uses an isolated cache because the deployed transformers version mutates a cache even under `use_cache=False`, which silently pollutes co-located measurements.

4.3 Mapper Ladder

We audit seven families under one interface. **Ridge (per-head)**: a 128×128 ridge map per student layer and head with Gram aggregation over variable-length calibration samples. **Affine (per-head)**: the same with mean centering and an intercept, fit in closed form over concatenated calibration samples. **Affine (per-layer)**: heads merged into rows, eight times fewer parameters. **Task-aware**: ridge initialization plus a bounded task fine-tune that mixes reconstruction with gold-answer cross-entropy (weight $\beta=0.3$) under a diagonal parameterization with a per-layer bounded gate. **RAT (residual-anchored translator)**: an architecture-derived map

$$W^{(s,h)} = [\pi(W_V^{S,h}) R \pi(W_V^{T,h})]^\top, \quad R = \arg \min_R \|E_T R - E_S\|_F^2 + \lambda \|R\|_F^2, \quad (1)$$

where $W_V^{T,h}$ and $W_V^{S,h}$ are the two models’ own value projections at the aligned layer pair, $\pi(\cdot)$ is the pseudo-inverse pullback from a head’s value space to the residual stream, and R aligns the two residual streams through the shared 151,936-token embedding matrices, which both Qwen3 models ship. A closed-form rank- r correction fits the calibration residual on top of the analytic core, a Hungarian matching on head covariance profiles replaces the identity head correspondence, and the sink position (position 0) is overwritten with the student’s calibration statistics because its translated value is a PPL catastrophe (Section 5.3). **Position-wise joint MLP**: for each student layer, a two-hidden-layer MLP with width 256 maps the concatenation of all aligned teacher layers and all KV heads at a position to all student heads, giving about 37M parameters across K and V, five times the per-head affine family; it is trained on the calibration split by Adam for 300 epochs. Keys follow the same construction on W_K . Every family is evaluated at calibration budgets 30 and 200 with disjoint evaluation samples ($n=30$ and $n=63$ respectively).

4.4 Oracle Probes

The probes are diagnostic interventions rather than universal upper bounds: they hand the student its *own* cache and mix in teacher content at a controlled budget. **Fraction probe**: $\alpha \hat{C}_S + (1 - \alpha) C_S$ for $\alpha \in \{0.25, 0.5, 0.75\}$ with $\alpha=0$ the student’s own cache. **Window probe**: translated content in only the bottom, middle, or top third of student layers, student content elsewhere. Because every configuration keeps the student’s own cache dominant or localized, a positive effect would demonstrate student-readable teacher-cache content, while degradation constrains—but does not exclude—nonlinear or jointly structured extraction paths. Probes are flagged non-deployable (they require the student’s own prefill) and exist only as measurement instruments.

4.5 Pairs, Data, and Protocol Discipline

Primary pair: Qwen3-4B (36 layers, hidden 2560) $\rightarrow$ Qwen3-1.7B (28 layers, hidden 2048); contrast pair: $\rightarrow$ Qwen3-0.6B (hidden 1024); scale-extended pair: Qwen3-8B (36 layers, hidden 4096) $\rightarrow$ 1.7B and 0.6B. All share the same tokenizer, vocabulary, $H=8$ KV heads, $d=128$, and RoPE base 10^6 . Every recorded run carries its git commit, mapper configuration, the protocol flags that define its regime, and per-sample scores; every number below is read directly from a recorded run. The accompanying repository holds 171 evaluation runs that record a metrics file; the rows below are a subset of them, and the deterministic configurations were re-run to check reproduction (Appendix A). We use multiple-choice QA deliberately: it gives a gold-probability control (the correct option’s log-probability is comparable across cache states), and it makes disjoint calibration/evaluation splits enforceable without leakage. We also run the audit on a 1K-token needle retrieval multiple-choice check (Section 5); task diversity beyond this format remains future work.

5 Results

5.1 H1 Passes: Mechanics Is Not the Failure Mode

Evidence tier: confirmatory. The identity control reproduces the student’s own prefill: per-sample logit cosine 0.99998, gold CHG +0.0001 $[-0.0006, +0.0009]$, $n=30$. The injection pipeline, cache construction, GQA head handling, and position accounting are therefore correct, and the gaps we report below are not attributable to student-space injection mechanics. The control holds across every configuration we report, including the cross-architecture and token-aligned runs (self-kv CHG within ± 0.001 ; Table 3). The control also holds through the disk-persistence path: an offline run that writes the cache and mapper parameters to disk and an online run that loads only the persisted store (no teacher in memory) produce identical metrics and identical per-sample gold probabilities (maximum absolute difference 0.000, $n=30$). Matching the two runs required persisting the mapper’s intercept, which an earlier implementation omitted; with that fixed, the cold-start path is exact rather than approximate.

The identity control also fixes the perplexity protocol used below. Injected caches are scored on the suffix conditioned on the cache, so the identity-injection PPL (21.2) is the student’s own suffix PPL and matches the native suffix distribution by construction. The student self-prefill row of Table 2 reports a different quantity—the native loss averaged over the full context and suffix in one pass—so its value (25.4) is not expected to equal the injected-cache PPL. The H1 identity criterion is defined on the answer-scoring logits, where the two paths agree to float tolerance.

5.2 No Strong-Student Mapper Establishes Replacement; the Weak-Student Result Is Inconclusive

Evidence tier: the strong-student ladder is confirmatory; the budget ladder, the mapper refits, the nonlinear families, and the 8B pair are frozen-gate extensions (Section 3.3).

Figure 2 and Table 2 give the landscape. Centering is the single biggest gain: per-head ridge (-0.296) drops to per-head affine (-0.138), roughly halving the deficit. The teacher and student value distributions differ in location and scale before they differ in content, which is exactly what the measured norm ratios say ($1.58\times$ for keys, $7.44\times$ for values). The calibration budget, by contrast, buys nothing. Holding the evaluation set fixed ($n=100$) and sweeping the calibration budget, affine mapping keeps CHG inside $[-0.257, -0.238]$: -0.251 at $c=10$, -0.249 at $c=30$, -0.238 at $c=60$, -0.257 at $c=100$, and -0.244 for the full calibration pool (budgets beyond the pool size saturate and reproduce the full-pool result exactly), a spread of 0.019. The affine mapper appears calibration-saturated within the tested budget: additional calibration examples in this range do not close the gap. Refitting the headline mapper on five independent calibration draws, with the evaluation set fixed, moves gold CHG within $[-0.292, -0.249]$ ($SD \approx 0.018$): the verdict is not an artifact of one calibration draw. Structure and function class change little either: per-layer merging (-0.275), task-aware fine-tuning (-0.216), RAT (-0.221 to -0.268), and a per-head MLP mapper with two GELU layers (CHG -0.39 to -0.26 at $c=30$ and -0.26 to -0.23 at $c=200$ across independent trainings) all sit inside the same band. Neither tested nonlinear family closes the gap. An over-parameterized position-wise joint mapper ($\sim 37M$ parameters) that reads all teacher layers and all KV heads also fails: across four independent trainings (the original run plus three seeded refits) its CHG spans -0.26 to -0.30 with accuracy 0.200, even though it reconstructs a fluent cache (PPL 29.6 versus 21.2 for the identity control). The strong student never approaches its own prefill (0.503); the best mapper reaches 0.365. Implementing the closest published strict-zero-reprefill design changes nothing: per-target-layer top- k source selection by held-out ridge reconstruction score with de-RoPE keys (the design of Heo et al.), at $k \in \{1, 3, 5\}$ and $c=200$ on the fixed evaluation set, degrades the strong student by -0.273 $[-0.380, -0.158]$, -0.308 $[-0.417, -0.200]$, and -0.291 $[-0.399, -0.192]$ respectively, with suffix perplexity rising from 78.5 to 6.9×10^4 as k grows. Aligning the two remaining implementation choices—concatenating the selected source layers and calibrating on FineWeb-Edu-style 1,024-token passages—does not rescue it: with 100 calibration sequences (the same token budget as $c=200 \times 512$; the reference uses 500) the CHG is -0.279 $[-0.410, -0.145]$, -0.223 $[-0.368, -0.084]$, and -0.254 $[-0.382, -0.122]$ for $k=1, 3, 5$, every interval below zero, and the identity control still reproduces the student exactly. The negative result therefore persists under the reference’s own combination and sequence-length regime; it remains not a strict reproduction only in that the reference calibrates on 500

FineWeb-Edu sequences where we use 100 sequences of a deterministically generated FineWeb-Edu-style corpus. Two aggregates summarize the strong-student result. Across the 128 translated configuration rows recorded under the audit protocol, the largest point estimate is +0.010, and it belongs to the weak-student pair; every strong-student row is negative. Three translated configurations clear the replacement margin of Section 3, and all three are near-chance students whose gains are indistinguishable from zero: Llama-3.2-1B both unaligned (+0.000 [−0.012, +0.012]) and aligned (−0.003 [−0.011, +0.005]), and Gemma-2-2B aligned (−0.007 [−0.018, +0.005]); their accuracies are 0.300 and 0.270 (Table 3).¹ Scaling the teacher to 8B under the same protocol does not change the verdict: self-injection remains an identity (logit cosine 0.99998) and the ridge mapper reaches CHG −0.224 [−0.320, −0.123] at $c=500$, so Failure to establish H2 persists at the teacher scale. The weak 0.6B student tells the opposite story. Affine at $c=200$ gives a near-parity point estimate +0.010 [−0.081, +0.102], and 8B→0.6B gives −0.023 [−0.104, +0.064]; both intervals include zero, but both lower bounds fall below the reporting replacement margin $-\epsilon = -0.02$, so non-inferiority is not established at these sample sizes. The two pairs fail that gate for different reasons: at the paired variance we observe for 4B→0.6B ($SD \approx 0.40$), establishing $\epsilon=0.02$ non-inferiority would require, conditional on the observed mean and variance, approximately 500 evaluation samples for the one-sided 95% interval lower bound to exceed −0.02 (roughly eight times the 63 we score). For 8B→0.6B, if the point estimate remained at −0.023, increasing the sample size alone would not satisfy the margin.

Evidence tier: the 1K-token retrieval task, the cross-architecture students, and the token-aligned variant are frozen-gate extensions (Section 3.3); neither replaces the primary pair. The failure is not specific to short commonsense contexts. On a 1K-token needle-in-a-haystack retrieval task (contexts of about 1024 tokens, $n=30$), where the teacher answers from a retrievable fact (0.733 accuracy versus the student’s 0.533), affine translation again fails: accuracy drops to 0.233 with gold CHG −0.285 [−0.534, −0.025], while identity injection remains exact (CHG +0.000 [−0.001, +0.001]). The deficit therefore tracks the translation regime rather than the context length or task format. We call this a 1K-token retrieval stress test rather than a long-context evaluation: ~1K tokens is short by current long-context standards. Extending the same task to a genuinely longer context was blocked by our host memory budget, and the initial “4K” configuration in fact fell back to the loader’s 1,024-token default—a plumbing bug we found and fixed while preparing this revision. With the fix, a true 4K-token retrieval run ($\approx 3.8K$ actual tokens, $n=10$) degrades the student by −0.596 [−0.929, −0.170] (accuracy 0.200 versus 0.900 for both teacher and student, suffix PPL 1.6×10^4), and a true 8K-token run ($\approx 7.6K$ actual tokens, $n=10$) by −0.577 [−0.932, −0.118] (accuracy 0.200, PPL 2.4×10^4): the deficit grows rather than shrinking with context length. We therefore report the long-context setting as further degradation at the sample sizes our 63 GB host budget permits, not as a parity result.

Cross-architecture checks relax the one-family scope (Table 3). Replacing the Qwen3 student with five models from other families—Llama-3.2-1B and 3B, Gemma-2-2B, Gemma-3-1B, and Qwen2.5-1.5B, spanning one to eight KV heads and head dimensions 64 to 256—mechanics still hold (self-kv CHG within ± 0.001) and translation never recovers the teacher’s 0.691 gold. Under the replacement gate of Section 3, one of the five (Llama-3.2-1B, near chance at 0.300 accuracy) is non-inferior but gains nothing, Gemma-2-2B sits at the margin, and the other three fall short (CHG −0.044 to −0.157; two of them with an interval excluding zero). Head groups are mean-pooled and head dimensions projected by the rectangular mapper; because the models tokenize independently, calibration pairs are matched position-by-position, so we read these as mechanism and replacement checks rather than fine-grained capability measurements. To address the tokenization confound directly, a token-aligned variant maps each student token to the teacher token with maximum character overlap before fitting and gives the same verdict on genuine cross-tokenizer pairs (Llama-3.2-3B: CHG −0.024 [−0.047, −0.004]; Gemma-3-1B: −0.073 [−0.134, −0.021]), so the negative result is not an artifact of independent tokenization; Qwen2.5 shares the Qwen tokenizer, making alignment a no-op there. Running the aligned variant for the two margin rows leaves the picture unchanged: Llama-3.2-1B gives −0.003 [−0.011, +0.005] and Gemma-2-2B −0.007 [−0.018, +0.005], so both still clear the replacement margin on near-chance students (accuracy 0.300 and 0.270) with gains indistinguishable from zero. Tokenization therefore does not explain the near-parity rows either.

¹Counted over the recorded run metrics by the repository’s aggregation script; “translated” excludes the identity, native-content, text-channel, and probe rows.

5.3 RAT Component Ablations Separate Fluency From Capability

RAT ablations attribute the failure inside the architecture-anchored design. Removing the sink override explodes suffix perplexity to 2.7×10^5 : the translated position-0 entry, which carries outsized attention mass, is a catastrophe unless overwritten with the student’s own statistics. Removing the analytic core and raising the correction rank to 32 *repairs fluency*: PPL drops to 23.7, within 2.5 of the identity control. Accuracy stays at 0.267. The analytic core’s min-norm pullback is harmful on real models, and more importantly, fluency recovery does not touch the answer deficit. The cache can be made to look right to the language-modeling objective without becoming right for the task. RAT’s role in the audit is to rule out the objection that the failure is an artifact of insufficiently architecture-aware translation.

5.4 H3: Teacher-Content Probes Do Not Improve on the Student

Evidence tier: the fraction and window probes on the primary pair are confirmatory; the octant sweep and the best-translator replication are frozen-gate extensions (Section 3.3). The probes test H3 without training anything (Figure 3). With the residual-anchored translator as the content source, gold probability falls from 0.503 to 0.380 at a 25% teacher fraction and to 0.249 at 75%; the point estimate is monotone in α , the intervals exclude zero from $\alpha=0.5$ onward ($-0.224 [-0.378, -0.082]$ and $-0.254 [-0.421, -0.079]$), and the mildest mixture is not separated from the student ($-0.123 [-0.261, +0.009]$). Replicating the probe with the best translator in the ladder (affine) preserves the direction and weakens the magnitude: $-0.086 [-0.182, +0.002]$ at $\alpha=0.25$, $-0.147 [-0.336, +0.019]$ at 0.5, and $-0.211 [-0.408, -0.022]$ at 0.75. No mixture at either source produces a gain whose interval excludes zero. Window injection localizes the damage: replacing the top third of layers is indistinguishable from the student (0.502 versus 0.503), while the middle and bottom thirds are strongly harmful (0.265 and 0.224). This is a sensitivity profile: the student’s behavior is substantially more sensitive to teacher-cache intervention in early and middle layers than in the top third. It does not by itself locate where the answer is routed; early and middle layers may also be more sensitive to off-manifold perturbations. An eight-way octant sweep ($n=30$) localizes the sensitive band: among the eight eighths of student layers, only layers 10–13 have a gold-probability interval excluding zero (CHG $-0.186 [-0.322, -0.060]$, gold 0.317 versus 0.503), while the top three eighths are statistically indistinguishable from the student (CHG -0.011 to $+0.020$). Two of those windows carry small positive point estimates ($+0.020$ and $+0.006$) whose intervals span zero; we report them explicitly rather than summarizing them away, because they are the only non-negative estimates the audit produces and they are consistent with noise in layers the student is insensitive to. The verdict is robust to translator quality in a narrow sense: neither the architecture-anchored translator nor the best mapper in the ladder produces content that yields a gain whose interval excludes zero. The family-wise bound below is a revision-added multiplicity analysis over this pre-specified confirmatory probe family. To state the probe result as a family-wise rather than a per-configuration claim, we take the maximum over the six confirmatory probe configurations (three mixture fractions and three layer windows, RAT source) and bootstrap that maximum over evaluation samples: the observed maximum is -0.001 and its 95% upper bound is $+0.009$. The bootstrap 95% upper estimate for the maximum gain across the confirmatory probe family is $+0.009$. Run over the full probe set instead—22 configurations, adding the best-translator replication and the eight-way octant sweep—the same computation gives $+0.075$.

Direct native-cache injection provides an additional negative control: injecting the teacher’s native (norm-rescaled) cache entries directly degrades the student to $-0.334 [-0.508, -0.154]$ ($n=30$), and the norm-rescaled native ablation replicates under the fixed-evaluation protocol ($-0.266 [-0.363, -0.168]$ at $c=30$ and $-0.251 [-0.351, -0.152]$ at $c=200$, $n=100$), so even teacher cache entries carried across layers without a channel map do not help under the fixed depth-ratio layer correspondence used throughout the audit (Appendix A): the harm is not an artifact of any particular channel map.

5.5 Perplexity and Accuracy Decouple

Figure 4 plots perplexity against accuracy for every method and run. The identity control and the student occupy the top-left regime. Translated caches populate a broad band, including a point with near-native perplexity and a 23-point accuracy deficit. Native injection sits at extreme perplexity (5.4×10^5) with low accuracy. Perplexity measures whether the cache permits fluent continuation; accuracy measures whether the

cache supports the task. The two dissociate, so perplexity cannot serve as evidence of task preservation—a caution that matters because perplexity is cheap and is often reported as a quality proxy.

5.6 Channel Asymmetry

Channel ablations at $c=30$ rank value-only (0.367) above key-only (0.300) above both (0.200) on accuracy, while perplexity orders them in reverse (88 versus 253 versus 1.8×10^7). Mapping keys repairs fluency and damages routing: the mapped keys steer attention to plausible but task-irrelevant positions. We report the asymmetry as a finding and leave the mechanism open. One hypothesis consistent with the asymmetry: mapped keys steer attention routing while mapped values carry content, so key errors misroute a fluent read while value errors corrupt it audibly.

6 Why Reconstruction Is Not Enough: An Architectural Analysis

We read this section as an architectural explanation of the audit’s verdicts, not an impossibility result: the argument is per-head and does not cover exploit paths that read the full multi-head, multi-layer cache (Section 8).

The audit’s verdicts fit the architecture. A cache entry is a linear projection $W_V h$ of a residual-stream state h . Translation must recover h from V , move it across the residual-space gap (2560 to 2048), and re-project. Two facts bound the recovery. First, the pullback $\pi(W_V)V$ returns only the row-space component of h ; the component orthogonal to W_V ’s rows, which can dominate a 2560-dimensional state observed through a 128-dimensional head, is invisible in that head’s value entry, so the per-head student value is underdetermined by its teacher counterpart. The student’s corresponding value $W_V^S R h$ depends on that invisible component, so the per-head student value is not determined by the per-head teacher value alone: the mapping task is intrinsically underdetermined, and a learned translator can at best track the conditional mean. The mapper-fitting view quantifies this: on held-out calibration contexts, the fitted per-head affine mapper recovers student key states (mean $R^2 = +0.81$) but only partially recovers value states (mean $R^2 = +0.32$): roughly two-thirds of the value variance is not linearly recoverable from the teacher’s per-head entry. The unit is a held-out calibration context: we fit on 20 contexts and score 10 disjoint held-out contexts; each yields a per-(layer,head) R^2 between predicted and true student states (pooled over token positions), and we report the mean over heads, layers, and the 10 contexts. The repository’s per-head ridge baseline agrees on the same held-out contexts (pooled $R^2 = +0.92$ for keys, $+0.29$ for values). Repeating the measurement for the position-wise joint MLP gives *higher* held-out reconstruction than the affine mapper—per-(layer,head) $R^2 = 0.854$ for keys and 0.481 for values, with training-split R^2 0.949 and 0.647 —and yet that mapper’s downstream change is worse (-0.27 versus -0.14). Better cache-state reconstruction does not buy task performance here, and the value channel stays only partially recoverable even on the fitting split. The closest published design fails on the same channel, and the verdict is stable across its two source-layer combinations. In our averaged variant, adding source layers makes reconstruction worse, not better—per-(layer,head) held-out R^2 falls from 0.822 to 0.787 to 0.762 for keys and from 0.317 to 0.248 to 0.182 for values as k goes from 1 to 3 to 5 (training-split value R^2 $0.430 \rightarrow 0.295 \rightarrow 0.222$), with suffix perplexity tracking the collapse ($78.5 \rightarrow 2629.6 \rightarrow 6.9 \times 10^4$). The aligned concatenation implementation inverts both diagnostics—held-out per-(layer,head) R^2 *rises* with k ($0.826 \rightarrow 0.858 \rightarrow 0.864$ for keys, $0.326 \rightarrow 0.419 \rightarrow 0.428$ for values) and suffix perplexity *falls* ($1.6 \times 10^5 \rightarrow 4.6 \times 10^4 \rightarrow 3.7 \times 10^3$)—yet the downstream CHG stays significantly negative throughout (-0.279 , -0.223 , -0.254). Better reconstruction and better fluency therefore do not move the task verdict in either direction, and the baseline’s instability is a property of the mapping on this pair rather than a case of good reconstruction with broken attention. Second, the teacher’s advantage over the student is consistent with being mediated by parameter-dependent computation that raw cache transfer does not reproduce. FFN layers carry the model’s factual associations [7], and the capability gap between Qwen3-4B and 1.7B is a gap in these parameters, which raw KV transfer does not explicitly transport. The oracle probes make the consequence concrete: even with the student’s own cache holding 75% of the mass, the teacher’s 25% contribution degrades the answer. MoT’s own design corroborates the analysis: its reported quality includes a target-side correction pathway that strict zero re-prefill disallows. Under a cold start the translator’s output is all the student gets, and that output shows no measurable teacher advantage in our audit.

7 Implications for Cache-Translation Research

Any future cache-translation evaluation should ship three instruments:

- an *identity control* that injects the student’s own cache through the full pipeline and must reproduce the student’s own prefill;
- a *gold-probability metric* with bootstrap intervals and permutation tests, reported as the change over the student’s own prefill;
- *oracle probes* that blend teacher-cache content into the student cache at controlled fractions and layer windows, probing exploitability within controlled interventions and without training anything.

Two gates apply to the metric: a replacement gate requires the bootstrap CI lower bound above a reporting tolerance $-\epsilon$ (the point estimate need not be non-negative), and a stricter gain gate requires the CI lower bound above zero. A pipeline delta over a student baseline is not evidence of translation, and a perplexity gain is not evidence of capability. Each instrument costs one evaluation run and composes with any serving stack.

For deployment, the weak student presents a near-parity regime rather than a demonstrated capability transfer. At 0.6B, translated injection is statistically indistinguishable from self-prefill (4B→0.6B: +0.010 [−0.081, +0.102]; 8B→0.6B: −0.023 [−0.104, +0.064]), but neither interval clears the reported $\epsilon=0.02$ replacement margin, so replacement-level service is not established, and at near-chance absolute accuracy (0.302 versus 0.25 random) this is not an endorsement. Top-layer injection is harmless, which means a system that must spend translation capacity can spend it there, though our probes also show it buys nothing there. Whether a capacity boundary separates a near-parity regime from failure is the question the 8B and 0.6B pairs begin to answer, and the audit protocol is built to test it. For strong students, cache translation is currently a net negative, and in this secondary comparison text compression loses less than the tested translation. On the fixed evaluation set the teacher’s summary of the context scores 0.440 (gold 0.404) against the student’s 0.520 (gold 0.512), a paired gap of −0.108 [−0.181, −0.037] ($n=100$), while the tested translation is worse still at −0.224 [−0.334, −0.117]. Both channels lose; the text channel loses less, and it carries no zero-re-prefill guarantee at all.

The survivable versions of the idea move work off the cache and onto the weights or the replay: allow a target-side correction pass the way MoT does, train student-side adapters, or translate through a shared latent space anchored by task vectors instead of per-entry caches. Each option gives up part of the cold-start guarantee, and each deserves a fresh identity control before its delta gets attributed to translation.

8 Limitations

Scope. Our scale pairs are all Qwen3, four in total, with $n=30$ to 100 evaluation samples per run; the interval widths reflect that sample size. The cross-family check covers four students from other families plus one same-family earlier generation, but it matches teacher and student positions after independent tokenization and one of its students is near chance (Table 3), so it establishes mechanics and gate-level replacement rather than capability.

Coverage of the alignment control. The token-aligned variant now covers all four cross-tokenizer students we report, including the two near-margin rows (Llama-3.2-1B and Gemma-2-2B); independent tokenization therefore does not explain either the clearly negative rows or the two near-parity rows. It remains untested for the same-tokenizer student (Qwen2.5), where alignment is a no-op by construction.

Probe family. The probes test translated content under linear blending, layer-window replacement, and an eight-way octant sweep, with native cache content as a content control. A nonlinear or jointly structured extraction path outside this family would not be detected. A positive probe result would have been decisive; our null result is not.

Translator capacity. H2 is scoped to translators trainable from the disjoint calibration budget we test (up to 200 examples). That set includes linear and nonlinear families and a 37M-parameter position-wise joint MLP, but it does not include a large-scale trained cross-attention translator, which is the one backbone from the related line that we neither reproduce nor rule out.

Variance. The mapper ladder uses a single calibration draw per configuration. Because a deterministic mapper is a function of that draw, we bound the draw variance directly: refitting the headline mapper on five independent draws with the evaluation set fixed moves gold CHG within $[-0.292, -0.249]$ (per-draw $SD \approx 0.018$), inside the per-run bootstrap intervals. The gradient-trained families behave differently: the per-head MLP spans -0.39 to -0.26 at $c=30$ and -0.26 to -0.23 at $c=200$ over five independent trainings, and the joint MLP spans -0.30 to -0.26 over four (Table 2); we report both as ranges, and the seeded refits reproduce given the seed.

What remains unmeasured. The cold-start equivalence rests on a single paired offline/online run rather than repeated pairs. The true 4K retrieval run is memory-limited to $n=10$, so its interval is wide even though it is significantly negative; the 1K row carries the context-length claim. The held-out reconstruction R^2 for both mappers comes from a single fit of 20 contexts and 10 disjoint held-out contexts, so it constrains rather than characterizes value-channel recoverability. Each top- k run now persists its selected layer table (`layer_mapping.json`); the selection is deterministic given the recorded configuration and calibration split. Reproduction artifacts cover the deterministic families, the fraction and octant probes, the 8B pairs, and two cross-architecture students; the 1K/4K retrieval rows and the aligned cross-architecture rows are not independently re-run. We report no end-to-end serving cost: the recorded pipeline measure comes from a non-deployment path that includes a measurement-only forward pass.

Prior work. We did not re-implement MoT’s replay-and-correction pipeline, and our replay diagnostic (Appendix C) shows only that a naive target-side re-read does not recover the strong student; it does not establish what MoT’s full pipeline does.

9 Conclusion

Cache translation promises to move a teacher’s work into a smaller model, and the audit separates that promise into three testable parts. The mechanism holds up: an identity control reproduces the student’s own prefill exactly. No tested strong-student map establishes replacement under the audit gate: seven mapper families, a calibration ladder, an architecture-anchored translator, and an implementation of the closest published strict-zero-reprefill design all leave the strong student below its own prefill, while the weak student’s near-parity point estimate remains inconclusive under the reporting margin. The map, moreover, is not where the problem appears to sit: no probe configuration delivers an interval-supported advantage, and a cache with near-native perplexity still fails the task. We state the negative narrowly: no configuration we test produces a gain whose interval excludes zero, the only configurations that clear the chosen replacement margin correspond to two near-chance students and show no measurable gain, and the probes leave a nonlinear or jointly structured extraction path untested rather than excluded. The evidence is consistent with an advantage that is weight-mediated rather than recoverable from raw KV states by the tested translators: recovering the teacher’s edge from raw KV states would take more than the tested translators provide (scope limits in Section 8). The audit framework is packaged so that the next cache-translation claim can be checked in one evaluation.

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A Reproducibility

Protocol timing. Three decisions were frozen at different points, and we separate them rather than describing the whole audit as pre-registered. The three-hypothesis framework and the probe family were fixed on 2026-08-29 (commits 0cc39ee, 54d7094, ca2fa89). The gate *rule*—non-inferiority against a small tolerance $-\epsilon$, with a separate gain gate, instead of requiring a non-negative point estimate—was fixed the same day (commit 5cbb4c3), before the fixed-evaluation-set calibration ladder of 2026-08-30/31. The numeric margin $\epsilon=0.02$ was fixed later, on 2026-09-12 (commit d46ba10), after those ladder runs; it is therefore a reporting choice, not a pre-registered constant. Two consequences matter for reading Table 2. The *gain* gate (95% CI lower bound above zero) does not depend on ϵ at all, and no configuration passes it. The *replacement* gate does depend on ϵ , so we report each row’s interval rather than only its gate label, and a reader who prefers a different margin can recompute the verdict from the interval. Confirmatory evaluation examples were never inspected under the audit metric before the framework freeze; the runs that came later are labelled as frozen-gate extensions in Section 3.3.

Statistics. Confidence intervals are percentile bootstrap intervals over per-sample scores with 10,000 resamples (paired: the same resampling indices are applied to both methods being compared). Permutation tests use 1,000 sign flips with the mean as the statistic and $p = (\#\text{extreme} + 1)/(n_{\text{perm}} + 1)$; no reported decision uses a p -value, and all gates are read from the intervals. The family-wise probe bound in Section 5 is a max-statistic bootstrap, $T^{*(b)} = \max_j \bar{\Delta}_j^{*(b)}$, over the same resampling scheme with a single set of resample indices shared by all configurations, so that sample-level correlation is preserved.

Gradient-trained rows. Table 2 reports a training range for the per-head MLP and the position-wise joint MLP because both are seed-dependent. The ranges and their worst-case interval bounds are:

Family	trainings	point-estimate range	worst CI upper bound
MLP per-head, $c=30$	5	−0.392 ... −0.257	−0.077
MLP per-head, $c=200$	5	−0.256 ... −0.226	−0.107
Joint MLP (position-wise), $c=30$	4	−0.302 ... −0.265	−0.110

The native-content control (Section 5) uses the same fixed depth-ratio layer correspondence as every other row, so it is a no-channel-map control rather than a translation-free injection.

Margin sensitivity. Because the numeric margin is a reporting choice, we report how the replacement verdict moves with it: at $\epsilon=0.01$ no translated configuration clears the margin; at the reported $\epsilon=0.02$, four rows (three distinct configurations, all two near-chance students) do; at $\epsilon=0.05$, seven rows (five distinct configurations, still confined to near-chance students and the smallest negative rows) do. The gain gate is unaffected by this choice.

Replacement margin	translated rows clearing	distinct configurations
$\epsilon = 0.01$	0	0
$\epsilon = 0.02$ (reported)	4	3
$\epsilon = 0.05$	7	5

Closest-published-baseline implementation. What we anchor to the reference design and what we chose ourselves:

Aspect	this audit implementation	anchored to the reference
Source-layer selection	per-target-layer top- k chosen by held-out ridge reconstruction score	yes (top- k cross-layer selection)
Selection/refit split	70% of the calibration contexts fit the selection ridge, 30% score it; the mapper is then refit on the full calibration split	not specified in the reference
Key mapping	de-RoPE keys (<code>de_rope_k</code>)	yes (de-RoPE keys)
Penalty	$\lambda = 10^{-3}$, K and V fitted separately	not specified
Candidate source layers	all teacher layers	not specified
Calibration corpus	FineWeb-Edu-style 1,024-token passages, replacing the audit-context calibration rows	FineWeb-Edu web text of the same length
Calibration volume	100 sequences, the same token budget as the $c=200 \times 512$ audit row	500 FineWeb-Edu sequences of 1,024 tokens
Source-layer combination	concatenation of the selected layers (aligned; our earlier averaged variant is also reported)	concatenation of the selected layers

Supplementary runs. The revision-added runs referenced above are recorded under the repository’s run directory: the closest-published-baseline grid (the `heo-topk` averaged runs and the `heo-concat-fwe` concatenation/FineWeb-Edu runs), the cold-start pair (the `persist` and `online` runs), the token-aligned margin rows (the `rect-align` runs for Llama-3.2-1B and Gemma-2-2B), the 4K retrieval run (`longctx4k`), the text-channel run with per-sample scores (`taskmix-summary`), and the joint-MLP reconstruction diagnostic (`joint_mlp_reconstruction`). The top- k selection code is `apcs.alignment.topk` and the concatenation mapper is `apcs.mapper.concat.ConcatRidgeMapper`; each run records its selected layer map in `layer_mapping.json`.

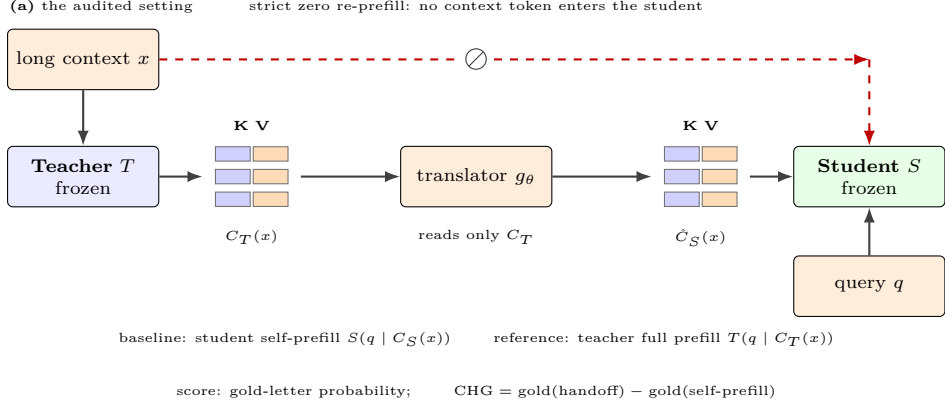
All numbers come from the accompanying repository’s recorded runs (`reports/runs/*/inject-eval/`), each carrying its git commit, the mapper configuration, the protocol flags that define its regime (`calib_eval_split`, `probe_mode`, `native_rescale`, `cross_token_align`), and per-sample scores. An independent reproduction of the deterministic configurations—ridge, affine, per-layer, task-aware, RAT, joint MLP, the calibration ladder, the fraction and octant probes, the 8B pairs, and the Llama-3.2-1B and Gemma-2-2B cross-architecture checks—matches the recorded values within 0.002; the gradient-trained per-head MLP and joint MLP are seed-dependent and reported as ranges (Table 2). The held-out reconstruction R^2 is produced by `scripts/measure_reconstruction_r2.py` (output under `reports/reconstruction_r2/`). The audit protocol is implemented in `apcs.inference.evaluator` with the flags `calib_eval_split`, `calib_shuffle`, `probe_mode`, and `native_rescale`; mapper families live in `apcs.mapper` (`math`, `joint`, `task_aware`, `rat`, the per-head MLP in `mlp`, the joint MLP in `joint_mlp`, and the cross-family rectangular map in `rect`); the 1K-token needle retrieval loader is `needle_mcqa` in `apcs.data.hf_dataset`; persistence and the cold-start online phase in `apcs.inference.kv_store`. Unit tests cover the audit counters, the store round-trip, and the mapper interface.

B Additional Measurements

The teacher/student KV norm ratios are $1.58\times$ (keys) and $7.44\times$ (values), measured over 30 calibration contexts per layer. Norm rescaling of the native cache repairs neither its perplexity (5.4×10^5) nor its accuracy, ruling out scale as the primary failure. The unified prompt renders choices once; the earlier protocol rendered them twice for the student baseline, which inflated the confidence-based deltas reported in our own preliminary sweeps and motivated the unified format in the audit.

C Append-on-Translated-Cache Replay Diagnostic

The audit holds strict zero re-prefill fixed, while MoT’s reported quality additionally includes a target-side replay with a Context Correction Loss. To probe how much a replay pass alone contributes, we add a diagnostic that re-reads the context through the student on top of the translated cache before scoring; it is explicitly non-deployable because it violates zero re-prefill. On Qwen3-4B $\rightarrow$ 1.7B (affine $c=30$, $n=100$), replay leaves accuracy at 0.290 and gold CHG at -0.232 [-0.333 , -0.136], statistically indistinguishable from the no-replay result (-0.249) and far below the student’s own prefill (0.520, CHG $+0.000$). Appending the context to a translated cache therefore does not recover the strong student. We do not test whether a trained correction pass would, and this diagnostic should not be read as evidence about MoT’s full pipeline: we do not implement MoT’s Context Correction Loss, so we cannot attribute its gains to replay alone.



(b) three questions that can fail independently, each with its own instrument and gate

Question	Instrument	Gate (fixed rule)	Outcome
H1 Injection mechanics: does the cache-injection pipeline preserve a valid student-space cache without loss?	Identity control: the student's own cache through the full pipeline	per-sample logit cosine = 1 †	pass: 0.99998
H2 Does any translator reach replacement level?	Seven mapper families; calibration 10–200 on one fixed evaluation set	95% CI lower bound > $-\epsilon$, $\epsilon=0.02^{\S}$ †‡	not met on the primary pair (best -0.138); two near-chance students (three aligned/unaligned configurations) clear it with no measurable gain
H3 Does the cache carry student-readable advantage?	Oracle probes: teacher fraction, layer window, octant; native content as control	some configuration's interval excludes 0 on the positive side †‡	none

Figure 1: The audited setting and the three gated questions. **(a)** A teacher prefills a long context x and stores its cache $C_T(x)$; a translator maps that stored cache into the student's cache space, and the student answers a short query q from the translated cache $\hat{C}_S(x)$. The red crossed path is the strict zero-re-prefill constraint: no context token enters the student's forward pass. The audit scores the handoff by the change in gold-letter probability against the student's own prefill (baseline), with the teacher's full prefill as the capability upper bound. **(b)** The three questions can fail independently (Section 1), so each carries its own instrument and gate; † marks gates fixed before the runs, and ‡ marks families and axes added while revising and evaluated under the same gates (Section 3.3).

Markers: † gate form fixed before the audit runs; ‡ families and axes added while revising under the same gates; § numeric replacement margin, selected later as a reporting tolerance (Appendix A).

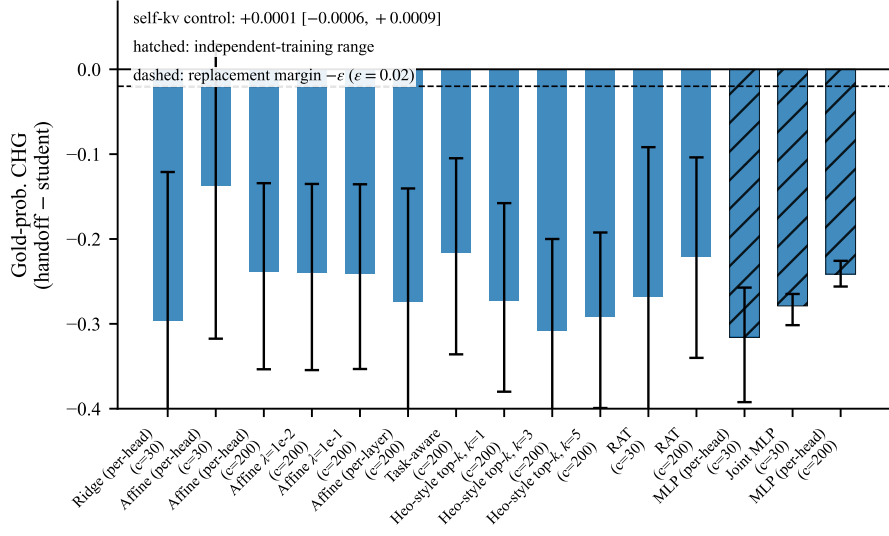


Figure 2: The mapper landscape for Qwen3-4B $\rightarrow$ 1.7B under the unified audit protocol: gold-probability CHG over the student’s own prefill, with 95% bootstrap CIs. All primary-pair point estimates are negative and no configuration passes the replacement gate; calibration budget moves affine mapping by only 0.005 on a fixed evaluation set; the architecture-anchored RAT, the gradient-trained MLP families (hatched bars span independent trainings), and the over-parameterized joint MLP sit inside the same band. The RAT, MLP, and joint-MLP bars are frozen-gate extensions; the remaining bars are confirmatory (Section 3.3).

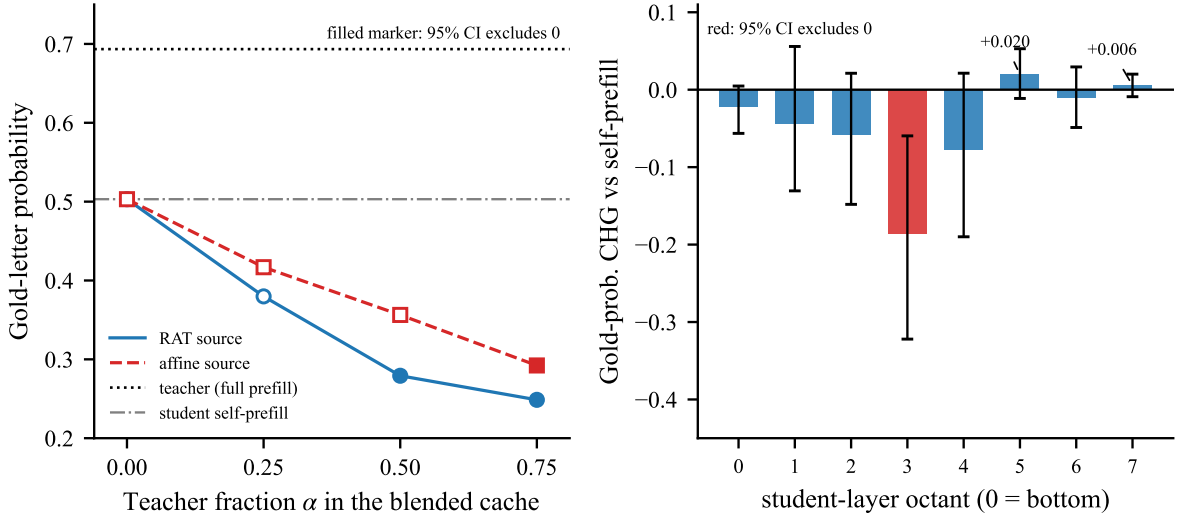


Figure 3: Oracle probes on Qwen3-4B $\rightarrow$ 1.7B ($n=30$; the fraction probe is confirmatory, the best-source replication and the octant sweep are frozen-gate extensions, Section 3.3). **Left:** gold-letter probability as translated teacher content replaces the student’s own cache, for two content sources. Filled markers mark mixture fractions whose paired CHG interval excludes zero: the decline separates from the student from $\alpha=0.5$ onward for the RAT source and only at $\alpha=0.75$ for the affine source, and no mixture at either source yields a gain whose interval excludes zero. **Right:** window injection per student-layer octant (affine source), plotted as gold-probability CHG with 95% CIs. Only the octant spanning layers 10–13 excludes zero; two top-layer octants carry small positive point estimates whose intervals span zero.

Table 2: Main audit results. Acc/gold/PPL: answer accuracy, gold-letter probability, and perplexity of the scoring suffix for the translated cache (kv-both). PPL entries for translated and identity caches score only the suffix conditioned on the injected cache; the student self-prefill row is the native loss over the full context and suffix in one pass and is not directly comparable (the identity-injection value is the student’s suffix PPL). CHG is the gold-probability change over the student’s own prefill with 95% bootstrap CIs. Pairs: Qwen3-4B→1.7B unless noted; rows marked 8B use the Qwen3-8B teacher. n_{eval} is the evaluation sample count (tail protocol rows use a fixed evaluation set of 100). All runs use disjoint calibration/evaluation splits and the unified prompt format. “Gate” applies the replacement gate of Section 3 ($\epsilon=0.02$): *pass* (CI lower bound above $-\epsilon$), *deg.* (CI upper bound below zero, i.e. significant degradation), *n.e.* (CI lower bound at or below $-\epsilon$ while the upper bound is at or above zero: neither non-inferiority nor degradation is established). A CI that contains zero therefore does not by itself imply *n.e.*; for the gradient-trained rows the CHG column is a training range rather than a single interval, with the number of trainings in the row label, and Appendix A lists each range with its worst-case interval bound. Calibration budgets above the pool size (200 examples once the fixed evaluation set is held out) saturate and reproduce the full-pool result. ‡ marks families and axes added while revising and evaluated under the same gates (Section 3.3); unmarked translation rows are confirmatory. † concatenates the selected source layers and calibrates on FineWeb-Edu-style 1,024-token passages, matching the reference’s combination and sequence length (100 sequences under our memory budget; the reference uses 500).

Configuration	Acc / Gold / PPL	Gold CHG [95% CI]	n_{eval}	Gate
Identity control (self-kv), 1.7B	0.500 / 0.503 / 21.2	+0.000 [−0.001, +0.001]	30	pass
Student self-prefill, 1.7B	0.500 / 0.503 / 25.4	—	30	—
Teacher (full prefill), 1.7B	0.767 / 0.693 / —	—	30	—
Native average, 1.7B	0.167 / 0.169 / 536082.5	−0.334 [−0.508, −0.154]	30	deg.
Ridge per-head, c=30, 1.7B	0.200 / 0.207 / 88.5	−0.296 [−0.479, −0.121]	30	deg.
Ridge per-head, c=500, 8B→1.7B‡	0.310 / 0.288 / 146.1	−0.224 [−0.320, −0.123]	100	deg.
Heo-style top- k ridge, $k=1$, c=200, 1.7B‡	0.240 / 0.238 / 78.5	−0.273 [−0.380, −0.158]	100	deg.
Heo-style top- k ridge, $k=3$, c=200, 1.7B‡	0.190 / 0.203 / 2629.6	−0.308 [−0.417, −0.200]	100	deg.
Heo-style top- k ridge, $k=5$, c=200, 1.7B‡	0.210 / 0.220 / 68966.1	−0.291 [−0.399, −0.192]	100	deg.
Heo-style top- k concat, FWE-1024, $k=1$, c=100‡	0.230 / 0.233 / 159446.7	−0.279 [−0.410, −0.145]	100	deg.
Heo-style top- k concat, FWE-1024, $k=3$, c=100‡	0.290 / 0.289 / 45921.5	−0.223 [−0.368, −0.084]	100	deg.
Heo-style top- k concat, FWE-1024, $k=5$, c=100‡	0.270 / 0.257 / 3677.7	−0.254 [−0.382, −0.122]	100	deg.
Affine per-head, c=30, 1.7B	0.367 / 0.365 / 46.1	−0.138 [−0.317, +0.029]	30	n.e.
Affine per-head, c=200, 1.7B	0.254 / 0.258 / 56.4	−0.239 [−0.354, −0.134]	63	deg.
Affine per-layer, c=200, 1.7B	0.206 / 0.222 / 81441.1	−0.275 [−0.407, −0.140]	63	deg.
Task-aware, c=200, 1.7B	0.270 / 0.280 / 142.1	−0.216 [−0.336, −0.105]	63	deg.
RAT, c=30, 1.7B‡	0.233 / 0.235 / 501.6	−0.268 [−0.440, −0.092]	30	deg.
RAT, c=200, 1.7B‡	0.238 / 0.276 / 1008.2	−0.221 [−0.340, −0.104]	63	deg.
RAT no-anchor rank-32, c=30, 1.7B‡	0.267 / 0.280 / 23.7	−0.223 [−0.367, −0.088]	30	deg.
MLP per-head, c=30 (5 trainings), 1.7B‡	0.187 / 0.187 / 80.2	−0.39 to −0.26	30	deg.
MLP per-head, c=200 (5 trainings), 1.7B‡	0.248 / 0.255 / 132.1	−0.26 to −0.23	63	deg.
Joint MLP (position-wise), c=30 (4 trainings), 1.7B‡	0.200 / 0.224 / 29.6	−0.30 to −0.26	30	deg.
Affine per-head, c=200, 4B→0.6B	0.302 / 0.283 / 71.8	+0.010 [−0.081, +0.102]	63	n.e.
Affine per-head, c=30, 8B→0.6B‡	0.270 / 0.269 / 99.0	−0.023 [−0.104, +0.064]	100	n.e.
RAT, c=200, 4B→0.6B‡	0.206 / 0.232 / 790.0	−0.041 [−0.120, +0.038]	63	n.e.
Affine per-head, c=30, 1K-token retrieval‡	0.233 / 0.232 / 6554.1	−0.285 [−0.534, −0.025]	30	deg.
Affine per-head, c=10, 4K-token retrieval (true 4K)‡	0.200 / 0.200 / 15855.1	−0.596 [−0.929, −0.170]	10	deg.
Affine per-head, c=10, 8K-token retrieval (true 8K)‡	0.200 / 0.200 / 24247.4	−0.577 [−0.932, −0.118]	10	deg.

Table 3: Cross-architecture robustness checks. Qwen3-4B teacher to non-Qwen and Qwen2.5 students under the within-family protocol (HellaSwag / ARC, $c=30$, $n=100$); identity injection passes in every row (self-kv CHG within ± 0.001). “KV h/d” gives the student’s KV heads and head dimension; the rectangular mapper mean-pools head groups and projects the head dimension. Rows marked “token-aligned” align teacher KV to student token positions by character overlap before fitting. No row recovers the teacher’s gold probability (0.691). “Gate” is the replacement gate of Section 3 ($\epsilon=0.02$, gold CHG); Gemma-2-2B sits at the margin (CI lower bound -0.020). Every row in this table is a frozen-gate extension (Section 3.3).

Student	KV h/d	Student acc / gold	Translated acc / gold	Gold CHG [95% CI]	Gate
Llama-3.2-1B	8/64	0.300 / 0.254	0.260 / 0.254	+0.000 [−0.012, +0.012]	pass
Gemma-2-2B	4/256	0.270 / 0.257	0.220 / 0.248	−0.009 [−0.020, +0.002]	fail
Llama-3.2-3B	8/128	0.390 / 0.281	0.200 / 0.237	−0.044 [−0.073, −0.013]	fail
Qwen2.5-1.5B	2/128	0.400 / 0.385	0.200 / 0.228	−0.157 [−0.209, −0.106]	fail
Gemma-3-1B	1/256	0.360 / 0.318	0.270 / 0.257	−0.061 [−0.135, +0.008]	fail
Llama-3.2-3B (token-aligned)	8/128	0.390 / 0.281	0.280 / 0.257	−0.024 [−0.047, −0.004]	fail
Gemma-3-1B (token-aligned)	1/256	0.360 / 0.318	0.250 / 0.245	−0.073 [−0.134, −0.021]	fail
Llama-3.2-1B (token-aligned)	8/64	0.300 / 0.254	0.310 / 0.251	−0.003 [−0.011, +0.005]	pass
Gemma-2-2B (token-aligned)	4/256	0.270 / 0.257	0.190 / 0.251	−0.007 [−0.018, +0.005]	pass

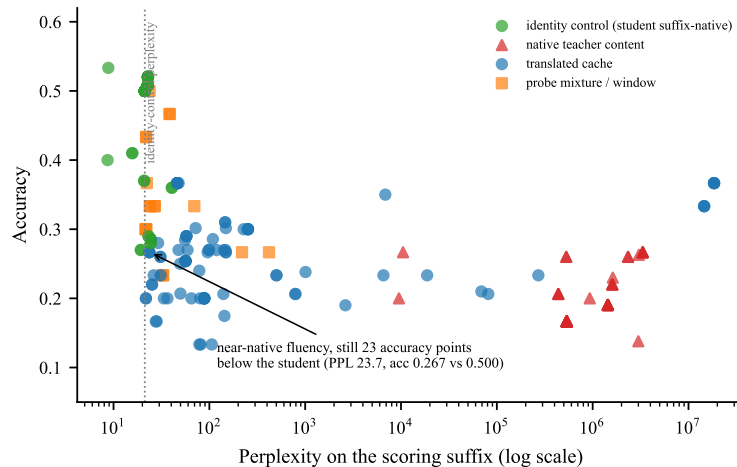


Figure 4: Perplexity over the scoring suffix versus accuracy for every method and run in the audit protocol (log scale; the superseded v1.1 runs are excluded). All points use the suffix-scoring protocol, so the identity control is the student’s suffix-native reference; the student’s full-context native loss (25.4, Table 2) is a different quantity and is not plotted. The identity control sits at low perplexity and high accuracy, while translated caches, probe mixtures, and native content spread across the plane—including a point with near-native perplexity (23.7) whose accuracy stays 23 points below the student’s 0.500. Perplexity does not order methods the way the task metric does.